Self-Reflection: Weather Utility App
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Description
I built a weather utility Android app from the ground up over five weeks. The application displays the current temperature, weather description, high, low, and next few hours forecast for Singapore. There is one screen for settings where the user can toggle between Celsius and Fahrenheit. I began by installing Android Studio and opening up the starter template, then moved onto Compose Layouts, Material Design 3, MVVM Architecture, and finally connected to the Open-Meteo API to get live weather data. Tests I completed were on my own OnePlus device, where the real time weather was correctly displayed.

Feelings
When I first loaded up the latest Android Studio version, it was awesome seeing a mobile application come to life right there on my desktop. Before long, I was wasting a whole afternoon frustrated and getting nowhere trying to get the virtual emulator to work properly. Then I got tired of playing with it and just plugged my physical device into the computer. It was so much simpler and clearer to see what was going on. Whenever something went wrong in my code (i.e. I had a compilation error), I'd see a red error message displayed. I'd click the red icon, read what the error message was saying, give it some thought on how I could potentially fix the issue and also reference my course notes. Material that the instructor provided was very well organised and provided me with support throughout the journey.

Evaluation
I believe my approach to solving issues was effective; that is I will attempt to resolve the issue on my own before requesting assistance. The course notes provided are well structured and included almost all material required for the course. The area where I have faced difficulties is in setting up the emulator which took a lot of time until I moved onto using a physical device. Also, sometimes I would make a simple mistake by not adding the internet permission under the right node in the manifest file.

Analysis
In my opinion, it's essential to work through an error independently first, since I can only truly grasp a problem by attempting to work it out myself, as well as recognise the areas where I'm lacking. If I simply used a tool that generates an answer for me right away, I will likely dismiss, or not value the answer given by AI even though it may have been provided in an instant. By implementing MVVM architecture, separating the UI from the data made my code more manageable. StateFlow automatically updated the screen when data changed, and Retrofit simplified the network request so I did not have to write HTTP code manually.

Conclusion
Through this assignment, I learned the basics of Kotlin, how to use Android Studio, and the overall process of building an Android app — from layout to architecture to connecting to a web API. I also practiced Git version control with regular commits.

Action Plan
If I continue developing this app, I would like to add a city search feature so users can check weather for different locations instead of only Singapore. I would also persist the temperature unit setting so it remains after the app restarts.

I also learned that breaking a project into small weekly commits made it much easier to track progress and roll back when something broke.